Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 08: EM waves at the surface of a conductor 2; Group velocity

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E and H in a good conductor

$$\vec{E} = \vec{E_0} e^{-\frac{z}{\delta}} e^{i\left(\frac{z}{\delta} - \omega t\right)}$$

Amplitude:

$$\vec{H} = \hat{k} \times \vec{E} \sqrt{\frac{\sigma}{\mu\omega}} e^{i\frac{\pi}{4}}$$

Direction:

$\vec{E}$, $\vec{H} \perp \vec{n}$, $\vec{k} // \vec{n}$

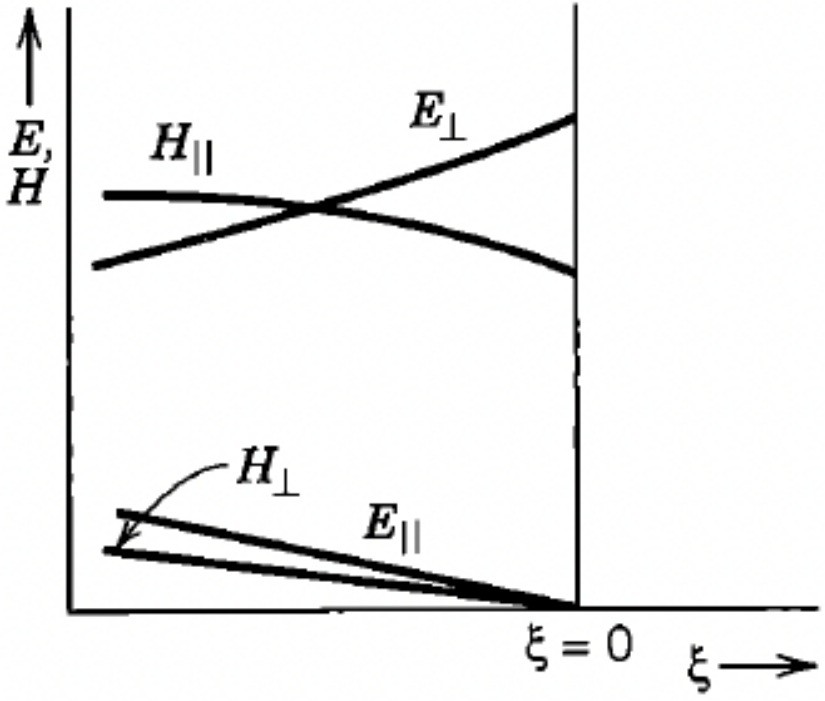
How about a perfect conductor?

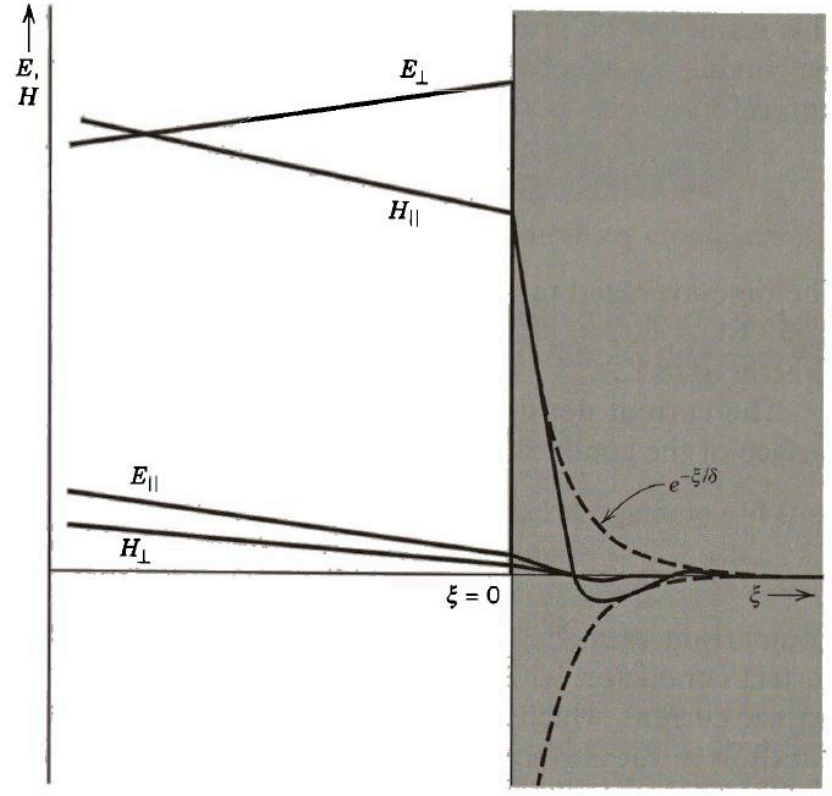
$\sigma \to \infty$

$\delta = \sqrt{\frac{2}{\mu \sigma \omega}}$ $\qquad \delta \to 0$

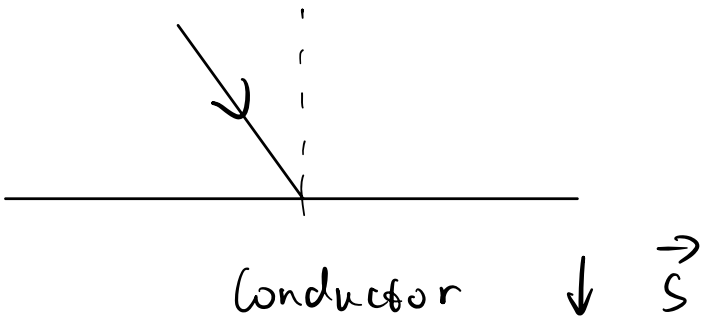
$\vec{E}, \vec{H} \to 0$

Perfect vs good conductor





Poynting Vector



$$\vec{S} = \vec{E} \times \vec{H} \qquad \hat{k} = \hat{z} \qquad \text{at } z=0$$

$$\langle S_{in} \rangle = \frac{1}{2} \mathrm{Re}\left(\hat{z} \cdot \left(\vec{E} \times \vec{H}^{*} \right) \right) = \frac{1}{2} \sqrt{\frac{\sigma}{2\pi\omega}} \left| \vec{E}_0 \right|^2$$

$$= \frac{1}{2} \sqrt{\frac{\pi\omega}{2\sigma}} \left| \vec{H}_0 \right|^2 = \frac{\pi\omega\delta}{4} \left| \vec{H}_0 \right|^2$$

Joule heating

$$\vec{J} = \sigma \vec{E} = \sigma \vec{E}_0 e^{-\frac{z}{\delta}} e^{i\left(\frac{z}{\delta} - \omega t\right)}$$

$$\langle P_J \rangle = \frac{1}{2} \vec{J} \cdot \vec{E}^* = \frac{1}{2} \sigma \vec{E} \cdot \vec{E}^*$$

$$= \frac{\sigma}{2} |\vec{E}_0|^2 e^{-\frac{2z}{\delta}}$$

$$= \frac{\sigma}{2} \frac{\omega\mu}{\sigma} e^{-\frac{2z}{\delta}} |\vec{H}_0|^2$$

$$= \frac{1}{\sigma\delta^2} e^{-\frac{2z}{\delta}} |\vec{H}_0|^2$$

Joule heating

$$\int_0^\infty \langle P_J \rangle \, dz = zp \, \frac{1}{2\delta^2} |\vec{E}_0|^2 \int_0^\infty e^{-\frac{2z}{\delta}} \, dz$$

$$= \frac{1}{2\delta^2} |\vec{E}_0|^2 \frac{\delta}{2}$$

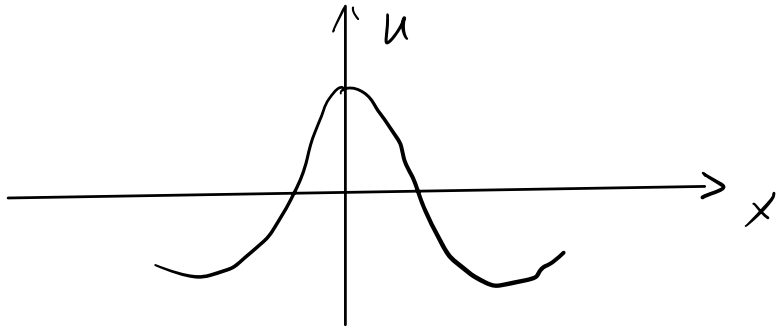
$$= \frac{1}{4\delta^2} |\vec{E}_0|^2 = \frac{1}{4} \sqrt{\frac{\omega\mu}{2}} |\vec{E}_0|^2 = \underline{\frac{1}{2} \sqrt{\frac{\omega\mu}{2}} |\vec{E}_0|^2}$$

$= \langle S_{in} \rangle$ All EM energy is converted to P_J.

General solutions of wave equations

- Monochromatic plane waves
- Dispersion relation

1D case



$$u(x,t) = \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} A(k)\, e^{i(kx - \omega t)}\, dk + \text{C.C.}$$

Assume $u(x,0) \neq 0$, $\frac{\partial u}{\partial t}(x,0) = 0$

$u(x,0)$ and $u(x,t)$ must be real.

$$A(k) = \frac{1}{2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u(x,0)\, e^{-ikx}\, dx$$

If $u(x,0)$ and $u(x,t)$ are complex

$$u(x,t) = \frac{1}{(2\pi)^{1/2}} \int dk \left(e^{ikx + i\omega t} + e^{i\vec{k}\cdot\vec{x} - i\omega t} \right) A(k)$$

Plane waves as the initial condition

$$u(x,0) = \cos k_0 x = \frac{1}{2}\left(e^{ik_0 x} + e^{-ik_0 x}\right)$$

$$A(k) = \frac{1}{4\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(e^{ik_0 x - ikx} + e^{-ik_0 x - ikx}\right) dx$$

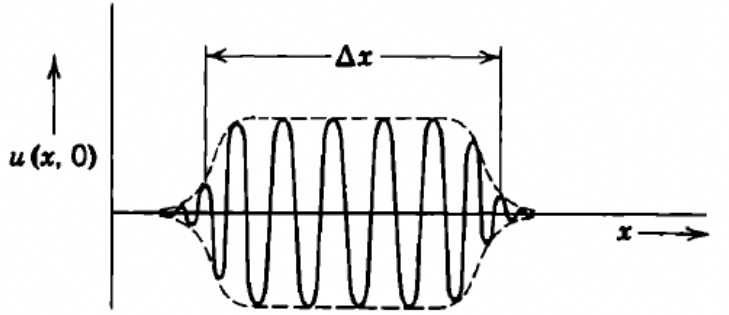
$$= \frac{\sqrt{2\pi}}{4}\left(\delta(k-k_0) + \delta(k+k_0)\right)$$

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\sqrt{2\pi}}{4}\left(\delta(k-k_0) + \delta(k+k_0)\right) e^{i(kx-\omega t)}\, dk \quad + \text{c.c.}$$

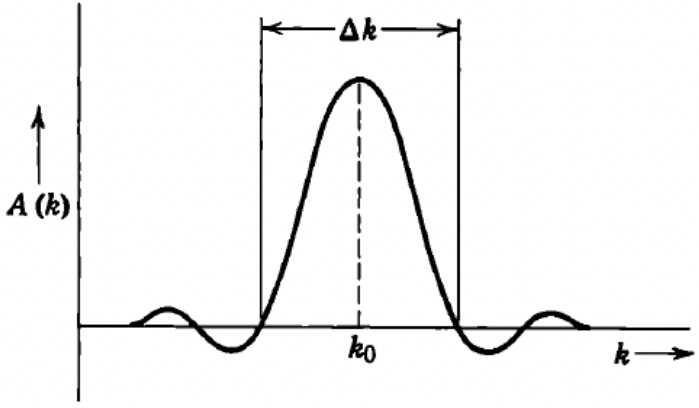
$$= \frac{1}{4}\left(e^{i(k_0 x - \omega t)} + e^{i(k_0 x + \omega t)}\right) \quad + \text{c.c.}$$

$$= \frac{1}{2}\cos(k_0 x - \omega t) + \frac{1}{2}\cos(k_0 x + \omega t)$$

Arbitrary initial condition



Δx is the mean square deviation of $|u(x,0)|^2$.



Δk is the mean square deviation of $|A(k)|^2$.

$$\Delta x \, \Delta k \geq \frac{1}{2}$$

Taylor expansion of dispersion relation

Δk is small. Δx is large.

- Assume the pulse is not too broad in its wave-number spectrum or its frequency depends weakly on wave number.

$$\frac{dw}{dk} << 1.$$

$$w = w(k) = w_0 + \left.\frac{dw}{dk}\right|_{k_0} (k - k_0) + \cdots$$

Group velocity; phase velocity

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k)\, e^{ikx - i\omega t}\, dk + c.c.$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k)\, e^{ikx - i\left(\omega_0 t + \left.\frac{d\omega}{dk}\right|_{k_0}(k-k_0)\right)t}\, dk + c.c.$$

$$= \frac{1}{\sqrt{2\pi}}\, e^{i\left(k_0 \left.\frac{d\omega}{dk}\right|_{k_0} - \omega_0\right)t} \int_{-\infty}^{\infty} A(k)\, e^{i\left(kx - \left.\frac{d\omega}{dk}\right|_{k_0} kt\right)}\, dk + c.c.$$

$$x' = x - \left.\frac{d\omega}{dk}\right|_{k_0} t$$

$$= \frac{1}{\sqrt{2\pi}}\, e^{i\left(k_0 \left.\frac{d\omega}{dk}\right|_{k_0} - \omega_0\right)t} \int_{-\infty}^{\infty} A(k)\, e^{ikx'}\, dk + c.c.$$

Group velocity; phase velocity

$$= e^{i(k_0 \frac{d\omega}{dk}\big|_{k_0} - \omega_0)t} u(x', 0) + c.c.$$

$$v_g = \frac{d\omega}{dk} \qquad x' = x - v_g t$$

$$v_p = \frac{\omega}{k}$$

Vacuum: $\omega = ck \Rightarrow v_g = c = v_p$

$$\omega = \frac{ck}{n(k)} \Rightarrow v_p = \frac{c}{n(k)}$$

$$v_g = \frac{d\omega}{dk} = \frac{c}{n} - \frac{ck\frac{dn}{dk}}{n^2} = \frac{c}{n} - \frac{ck}{n^2} v_g \frac{dn}{d\omega} \Rightarrow v_g = \frac{c}{n + \omega \frac{dn}{d\omega}}$$

Group and phase velocity of lights

Normal dispersion

$v_p < c$, $\frac{dn}{d\omega} > 0$

$v_g < v_p < c$

Abnormal dispersion .

$\frac{dn}{d\omega} < 0$ $\quad v_g > v_p$

